CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 11: Geometric Progression

Topic-wise short notes with solved examples

A geometric progression is a sequence in which each term is a fixed multiple of the one before it. This chapter covers its general term, the sum of its terms and the geometric mean.

1 What a G.P. is

- A **geometric progression (G.P.)** is a sequence of non-zero numbers in which the ratio of any term to the term before it is constant.
- The constant is the **common ratio** r, and the first term is a.

$$r = \frac{t_2}{t_1} = \frac{t_3}{t_2} = \dots \qquad \text{General G.P.: } a,\ ar,\ ar^2,\ ar^3,\ \dots$$

- r can be a fraction or a negative number, but it cannot be zero. No term of a G.P. is zero.
- To test for a G.P., check that the ratios of consecutive terms are all equal.

Example 1.1 Which of these are G.P.s? (i) $3, 6, 12, 24, \dots$ (ii) $2, 4, 6, 8, \dots$ (iii) $81, -27, 9, -3, \dots$

SOLUTION

(i) Ratios: $2, 2, 2$. Equal.

(ii) Ratios: $2, \frac{3}{2}, \frac{4}{3}$. Not equal (this is an A.P.).

(iii) Ratios: $-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}$. Equal.

(i) is a G.P. with $r = 2$; (iii) is a G.P. with $r = -\frac{1}{3}$; (ii) is not a G.P.

2 The general term

$$t_n = a\,r^{\,n-1}$$

- If the last term is l, the nth term from the end is $\frac{l}{r^{\,n-1}}$.
- Dividing one term by another removes a: $\frac{t_m}{t_n} = r^{\,m-n}$.

Example 2.1 Find the 8th term of the G.P. $3, 6, 12, \dots$

SOLUTION

$a = 3, r = 2$

$t_8 = 3 \times 2^7 = 3 \times 128$

$t_8 = 384$

Example 2.2 Which term of the G.P. $2, 6, 18, \ldots$ is 1458?

SOLUTION

$2 \times 3^{n-1} = 1458$, so $3^{n-1} = 729 = 3^6$

$n - 1 = 6$

The 7th term.

Example 2.3 The 4th term of a G.P. is 24 and its 7th term is 192. Find the G.P.

SOLUTION

$ar^3 = 24$ and $ar^6 = 192$

Divide: $r^3 = 8$, so $r = 2$ and $a = \dfrac{24}{8} = 3$

The G.P. is $3, 6, 12, 24, \ldots$

3 Sum of n terms

$$S_n = \frac{a(r^n - 1)}{r - 1} \text{ when } r > 1 \qquad S_n = \frac{a(1 - r^n)}{1 - r} \text{ when } r < 1$$

- Both forms give the same value; choosing by the size of r keeps the numbers positive.
- When $r = 1$ every term equals a, so $S_n = na$.

Example 3.1 Find the sum of the first 8 terms of the G.P. $1, 2, 4, \ldots$

SOLUTION

$a = 1, r = 2, n = 8$

$$S_8 = \frac{1 \times (2^8 - 1)}{2 - 1} = 256 - 1$$

$S_8 = 255$

Example 3.2 Find the sum of the first 6 terms of the G.P. $81, 27, 9, \ldots$

SOLUTION

$a = 81, r = \dfrac{1}{3}, n = 6$

$$S_6 = \frac{81\left(1 - \frac{1}{729}\right)}{1 - \frac{1}{3}} = 81 \times \frac{728}{729} \times \frac{3}{2} = \frac{728}{9} \times \frac{3}{2}$$

$S_6 = \dfrac{364}{3} = 121\frac{1}{3}$

Example 3.3 How many terms of the G.P. $3, 6, 12, \ldots$ are needed to give a sum of 381?

SOLUTION

$\frac{3(2^n - 1)}{2 - 1} = 381$, so $2^n - 1 = 127$ and $2^n = 128 = 2^7$

7 terms.

4 Geometric mean

- If a, G, b are in G.P., then G is the **geometric mean** of a and b.

$$G^2 = ab \qquad G = \sqrt{ab}$$

- Three non-zero numbers a, b, c are in G.P. if and only if $b^2 = ac$.

Example 4.1 Find the geometric mean of 4 and 25, and the value of x if x, 6, 12 are in G.P.

SOLUTION

$G = \sqrt{4 \times 25} = \sqrt{100} = 10$

$6^2 = 12x$, so $x = 3$

Geometric mean 10; $x = 3$

5 Choosing terms conveniently

- Three numbers in G.P.: take them as $\frac{a}{r}$, a, ar. Their product is a^3.

Example 5.1 The product of three numbers in G.P. is 216 and their sum is 26. Find the numbers.

SOLUTION

Let the numbers be $\frac{a}{r}$, a, ar. Product: $a^3 = 216$, so $a = 6$.

Sum: $\frac{6}{r} + 6 + 6r = 26$, so $6r^2 - 20r + 6 = 0$ and $3r^2 - 10r + 3 = 0$

$(3r - 1)(r - 3) = 0$, so $r = 3$ or $r = \frac{1}{3}$

The numbers are 2, 6 and 18.

6 A.P. and G.P. compared

	A.P.	G.P.
Constant	Difference d	Ratio r
General term	$a + (n-1)d$	$a\,r^{n-1}$
Three terms	$a-d,\ a,\ a+d$	$\frac{a}{r},\ a,\ ar$
Condition	$2b = a + c$	$b^2 = ac$
Mean of a and b	$\frac{a+b}{2}$	$\sqrt{ab}$

7 Common mistakes

- Writing $t_n = a\,r^n$. The power is $n-1$.
- Finding r as $\frac{t_1}{t_2}$. It is $\frac{t_2}{t_1}$.
- Sign errors with a negative r: odd powers are negative, even powers positive.
- Using the A.P. sum formula for a G.P.

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